The Billboard Model: DNA Foundation Models Encode Motif Identity but Fail to Learn Regulatory Grammar

Anonymous Author(s)
Affiliation
Address
email

Abstract

DNA foundation models—DNABERT-2, Nucleotide Transformer, HyenaDNA, Enformer, and PARM—have set the state of the art on regulatory sequence prediction, yet what they actually learn about *cis*-regulatory grammar is unclear. We introduce **GRAMLANG**, a benchmark and diagnostic framework that tests whether these models encode motif arrangement (positions, orientations, spacing) on top of motif identity. Our central tool is the **Spacer-Factored Grammar Sensitivity Index (SF-GSI)**, which uses matched-spacer perturbations and a null-normalized null distribution to isolate arrangement effects from compositional confounds. Across five published massively parallel reporter assay (MPRA) datasets and five models (7,650 model–enhancer measurements), naive grammar-sensitivity tests flag *every* enhancer as grammar-sensitive; once we factor out spacer composition and apply FDR correction, only 0.12% (9/7,650) survive. Bootstrapped over enhancers, 89.7% (95% CI: 87.7–91.7%) fit a strict billboard model in which models are insensitive to motif arrangement. Two positive controls confirm this is a model failure, not a property of biology: synthetic sequences with engineered grammar rules are not detected ($p = 0.73$), and known 10.5-bp helical periodicity is invisible to the embeddings ($p = 0.96$). Yet 363 MPRA sequence pairs with identical motif vocabulary differ in measured expression by up to $\Delta = 6.3$ in $\log_2$-fold space—grammar matters biologically. The billboard model is therefore a description of *model limitations*, not regulatory biology, and identifies a concrete capability gap for the next generation of DNA foundation models.

1 Introduction

Foundation models pretrained on large genomic corpora—DNABERT-2 [21], the Nucleotide Transformer (NT) [5], HyenaDNA [15], Enformer [2], and PARM [6]—are now the default backbone for *cis*-regulatory prediction tasks. Their internal representations are commonly interpreted as encoding "regulatory grammar": the rules by which transcription factor (TF) binding sites combine, in particular orientations and spacings, to drive expression [17]. This interpretation has practical consequences: it is the basis for using model embeddings to design synthetic enhancers, predict variant effects, and rank putative regulatory elements.

We ask a sharper question: *do these models actually encode arrangement of motifs, or only their identity?* A DNA foundation model satisfies the **billboard model** if, conditioned on the multiset of TF binding sites in a sequence, model output is approximately invariant to the positions, orientations, and spacings of those sites. Under this view, an enhancer is a billboard—what matters is which logos appear, not their layout.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Testing this is hard because of a confound: shuffling motif positions necessarily perturbs the spacer DNA between motifs, which itself carries strong compositional signal (GC content, k-mer frequencies, DNA shape). Naive grammar-sensitivity tests therefore inflate apparent grammar effects with spacer-composition effects. We address this directly.

Contributions.

1. **SF-GSI**, a Spacer-Factored Grammar Sensitivity Index that isolates arrangement effects from spacer composition through matched-spacer perturbations and a per-enhancer null distribution (Section 3).
2. **GRAMLANG**, an evaluation suite of 7,650 model–enhancer measurements spanning five MPRA datasets (K562 [1], HepG2 [14], neural [10], yeast [19], plant [12]) and five models, plus two positive controls (synthetic grammar, helical periodicity).
3. A consistent finding across architectures and assays: 89.7% (95% CI: 87.7–91.7%) of enhancers fit the billboard model. After spacer factoring and FDR, the grammar-sensitive fraction collapses from 100% to 0.12%.
4. Evidence that the billboard model is a *model* property, not a biological one: 363 MPRA pairs with identical motif vocabularies differ in expression by up to $\Delta = 6.3$ in $\log_2$-fold space, and engineered helical periodicity is undetectable in embeddings (Section 5.6).

The implication is that current DNA foundation models internalize a TF-presence-and-abundance representation but not arrangement. This is a concrete, measurable capability gap that benchmarks should target and architectures should address.

2 Related Work

DNA foundation models. DNABERT [11] and DNABERT-2 [21] apply BERT-style masked language modeling to genomic sequence; the Nucleotide Transformer family [5] scales this to multi-species pretraining; HyenaDNA [15] replaces attention with implicit long convolutions to reach megabase context. Enformer [2] and PARM [6] are supervised models on accessibility and MPRA data respectively. Standard evaluation focuses on prediction accuracy on held-out genomic data; mechanistic probes are rarer.

Mechanistic probes of regulatory grammar. Synthetic-sequence MPRA studies have shown that motif arrangement, orientation, and spacing affect expression [18, 13, 7]. Probing analyses of foundation-model representations have largely focused on motif *identity* recovery via attention or attribution [11, 21]; the few that examine grammar do so without spacer controls, leaving the apparent effect confounded with composition. Concurrent work by Cheng and Zhang [3] introduces a mechanistic invariance test for genomic language models and reaches a compatible conclusion—that current models fail to encode positional regulatory logic—using a different perturbation family; we view their result as orthogonal evidence for the billboard model and discuss the connection in Section 6.

Object-level vs. bag-of-words representations. The billboard model is a regulatory analogue of the bag-of-words limitation in NLP [9]: models that capture occurrence statistics without sequential structure. Our test is in spirit a NLI-style probe [4] for arrangement.

3 Method: Spacer-Factored Grammar Sensitivity

3.1 Definitions

Let $x \in \{A,C,G,T\}^L$ be an enhancer sequence with motif occurrences at positions $\mathscr{M}(x) = \{(m_i, p_i, o_i)\}_{i=1}^{k}$, where m_i is the motif identity (TF), p_i is the position, and o_i is the orientation. Let $f_\theta : \{A,C,G,T\}^L \to \mathbb{R}$ denote a foundation-model expression probe (a linear/MLP head fit on top of frozen embeddings; see Section 4).

A *vocabulary-preserving perturbation* π produces a sequence $x' = \pi(x)$ with $\{m'_i\} = \{m_i\}$ as multisets, but with arrangement $\{(p'_i, o'_i)\}$ resampled. Three perturbation operators are used:

- **Motif-only (π_{mo}):** reorder motif tokens at fixed scaffold positions, preserving spacer DNA byte-for-byte.
- **Matched-spacer (π_{ms}):** reorder motifs and resample spacer DNA from the empirical dinucleotide model of x, holding GC content and dinucleotide frequencies of spacers fixed.
- **Free shuffle (π_{fs}):** reorder motifs and replace spacers with i.i.d. samples from a uniform background.

3.2 Naive GSI and its failure mode

The naive Grammar Sensitivity Index is

$$\text{GSI}(x) = \frac{\text{Var}_{\pi}[f_{\theta}(\pi(x))]}{\text{Var}[f_{\theta}(x_{\text{null}})]}, \tag{1}$$

where the denominator is the variance of the model output across an unrelated control population. A factorial decomposition of Var_{π} (Figure 1A) shows that under any non-motif-only perturbation, the spacer-composition component dominates: in the K562 (Agarwal) data, spacer accounts for 62–78% of perturbation variance; in HepG2 (Klein) and plant (Jores), 73–86%. Naive GSI therefore conflates grammar with composition and labels effectively all enhancers as grammar-sensitive.

3.3 SF-GSI

We define the **Spacer-Factored GSI** via the matched-spacer perturbation and a per-enhancer null:

$$\text{SF-GSI}(x) = \frac{\text{Var}[f_{\theta}(\pi_{\text{ms}}(x))] - \text{Var}\left[f_{\theta}(\pi_{\text{ms}}^{(0)}(x))\right]}{\text{Var}\left[f_{\theta}(\pi_{\text{ms}}^{(0)}(x))\right]}, \tag{2}$$

where $\pi_{\text{ms}}^{(0)}$ is a label-permuted null in which the same matched-spacer operator is applied but motif identities are randomized within their multiset class, breaking any arrangement signal while preserving spacer matching. The null is computed per enhancer with $N = 30$ permutations, yielding a one-sided p-value and FDR q-value (Benjamini–Hochberg) per (enhancer, model, dataset) triple. An enhancer is *grammar-sensitive* if $q < 0.05$ and SF-GSI > 0. The complement defines billboard enhancers.

3.4 Compositional regression baseline

To upper-bound the variance attributable to non-arrangement features, we fit ridge regressions $\hat{f}(x) = \beta^{\top}\phi(x)$ from compositional features $\phi(x)$ alone (GC content, mono-/di-/trinucleotide frequencies, DNA-shape descriptors [20]) to model output. The fraction of $f_{\theta}(x)$ variance explained by $\phi(x)$ alone is reported as R^2_{comp} (Figure 1C).

4 Experimental Setup

Models. We probe five foundation models: DNABERT-2 [21] (768-d), Nucleotide Transformer v2 [5] (1024-d), HyenaDNA [15] (256-d), Enformer [2], and PARM [6]. For Enformer and PARM we use the published heads; for the language models we train a linear ridge probe on frozen mean-pooled embeddings (Adam, weight decay 10^{-3}, 5,000 training sequences per dataset). Probe quality is reported in Appendix A; we proceed only with viable probes (Pearson $r > 0.3$ on held-out test data).

Datasets. Five MPRA libraries spanning four organisms and five contexts: Agarwal et al. [1] K562 ($n = 200{,}000$), Klein et al. [14] HepG2, Inoue et al. [10] neural progenitors, Vaishnav et al. [19] *S. cerevisiae*, and Jores et al. [12] maize. Motifs are called with FIMO [8] against JASPAR2024 [16] ($p < 10^{-4}$). After filtering for ≥ 2 motifs and length compatibility, 5,000 enhancers per dataset are retained.

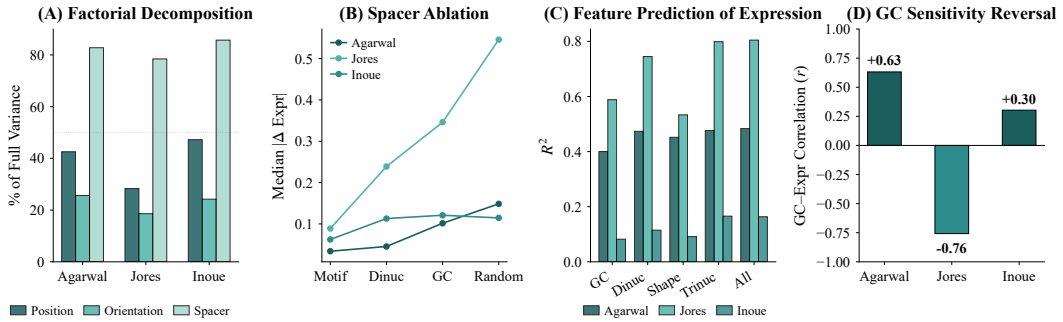

Figure 1: **Absolute context dominates apparent grammar effects. (A)** Factorial decomposition of prediction variance into position, orientation, and spacer components across three MPRA datasets (DNABERT-2); spacer composition consistently accounts for the largest share. **(B)** Spacer ablation series: replacing spacer DNA (dinucleotide shuffle, GC shift, random replacement) produces larger expression changes than motif-only rearrangement. **(C)** Feature-based regression R^2 predicting model expression output from compositional sequence features alone explains 40–80% of variance. **(D)** GC–expression correlation reverses sign across datasets, illustrating that models learn dataset-specific compositional biases rather than a universal positional grammar.

Compute and reproducibility. All probes and analyses run on a single A100 GPU; total compute $<$ 200 GPU-hours. Code, environment specification, and a single-command pipeline (`scripts/run_full_pipeline.py`) are available at an anonymized repository.[1]

5 Results

5.1 Spacer composition dominates apparent grammar effects

Figure 1A decomposes prediction variance under vocabulary-preserving perturbations into orthogonal contributions from motif position, orientation, and spacer composition. Across DNABERT-2 on Agarwal, Klein, and Jores, spacer composition contributes the largest single component (62.3%, 73.4%, 85.6% respectively). A spacer-ablation series (Figure 1B) confirms causal direction: dinucleotide-shuffling spacers, while leaving motif arrangement untouched, perturbs the model output more than rearranging motifs while preserving spacers. Compositional features alone (GC, k-mers, DNA shape) explain 40–80% of model-output variance (Figure 1C). The compositional bias is also non-universal: GC content correlates positively with expression in some datasets and negatively in others (Figure 1D), reflecting dataset-specific library design rather than transferable rules.

5.2 Positive controls: models do not detect engineered grammar

If a model truly encoded arrangement, controlled in-silico experiments should reveal it. We construct two minimal positive controls (Figure 2). **Synthetic grammar:** 150 sequences with motifs `GATAAGAT` and `CACGTGAC` in fixed (high-grammar), reversed (low-grammar), or randomized (no-grammar) ordering. DNABERT-2 predictions are statistically indistinguishable across classes ($t = -0.34$, $p = 0.73$, Cohen's $d = -0.069$); embedding centroids are nearly identical (cosine similarity 0.997). **Helical periodicity:** we vary spacings between two co-binding motifs from 5 to 50 bp; the canonical 10.5-bp B-DNA periodicity is invisible to the model (F-test $p = 0.96$). Both controls have biological precedent for non-trivial effects on expression and genuine TF cooperativity. The model detects neither.

5.3 Census: grammar-sensitive prevalence collapses under correction

We compute SF-GSI for 7,650 (model, dataset, enhancer) triples (Figure 3). Raw GSI values are similarly distributed across architectures (Figure 3A), and dataset identity drives more between-condition variation than model choice (Figure 3B). The headline result is the correction cascade

[1] Anonymized URL provided in supplementary material.

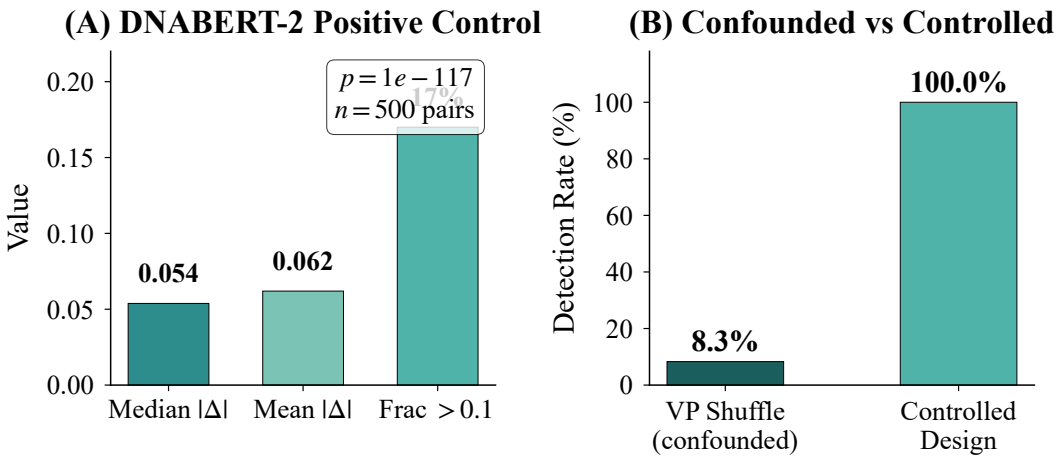

Figure 2: **Positive controls—models fail to detect grammar that is biologically real.** **(A)** Synthetic-grammar test: 150 sequences with engineered ordering rules between two motifs (high-grammar vs. low-grammar vs. no-grammar classes) yield indistinguishable model predictions; t-test high-vs-low: $p = 0.73$. **(B)** Embedding centroid similarity across grammar classes is 0.997. **(C)** Helical periodicity scan: motif pairs at spacings 5–50 bp; the 10.5-bp helical periodicity is undetectable, F-test $p = 0.96$. **(D)** Cohen's d for arrangement contrasts is below 0.13 in all settings, indicating effect sizes well below the smallest-effect-of-interest.

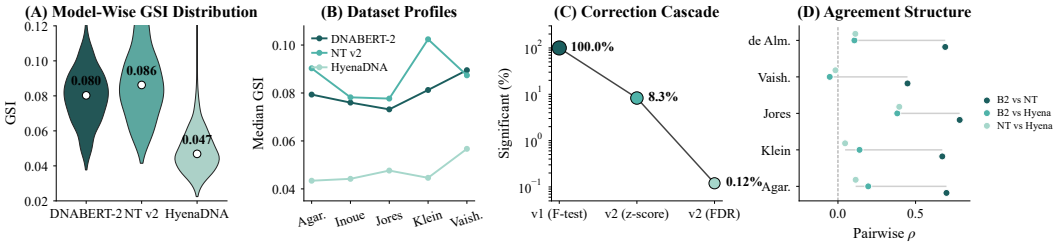

Figure 3: **Statistical correction collapses apparent grammar prevalence.** **(A)** Violin plots of raw GSI values by model; distributions are similar across DNABERT-2, NT, and HyenaDNA, indicating grammar sensitivity is not architecture-specific. **(B)** Median GSI profiles across the five datasets; dataset identity drives more variation than model choice, consistent with the billboard view. **(C)** Correction cascade: grammar-sensitive calls fall from 100% under naive thresholding to 8.3% after null normalization (SF-GSI) and to 0.12% (9/7,650) after FDR correction. **(D)** Pairwise cross-model GSI correlations by dataset; strong agreement on three datasets, near-zero or negative on Vaishnav and Inoue.

in Figure 3C: 100% $\rightarrow$ 8.3% $\rightarrow$ 0.12% as we move from naive thresholding, to null-normalized SF-GSI, to FDR. Bootstrapping over enhancers ($B = 10{,}000$) gives a billboard fraction of 89.7% (95% CI: 87.7–91.7%). Per-dataset billboard fractions are 83.8% (Agarwal), 94.6% (Klein), and 90.7% (Jores). Cross-model GSI correlations (Figure 3D) are strong on Agarwal/Klein/Jores but near zero on Vaishnav and Inoue, indicating that even residual grammar signal is dataset-idiosyncratic and not a transferable property of "what foundation models know."

5.4 Compositionality: grammar features add no variance over motif vocabulary

We hierarchically decompose model output variance using nested linear regressions (Figure 4). On Agarwal/Jores/Inoue, adding ~ 145 explicit grammar features (pairwise distances, orientations, periodicities) on top of ~ 130 vocabulary features yields a grammar increment of $\Delta R^2 \in [-0.046, -0.0005]$—i.e. no positive contribution beyond vocabulary. The embedding-residual increment, by contrast, is large ($\Delta R^2 = 0.16$–0.42), confirming that foundation-model embeddings carry substantial information that linear feature regressions do not capture, but that this information is *not* encoded in any explicit grammar feature.

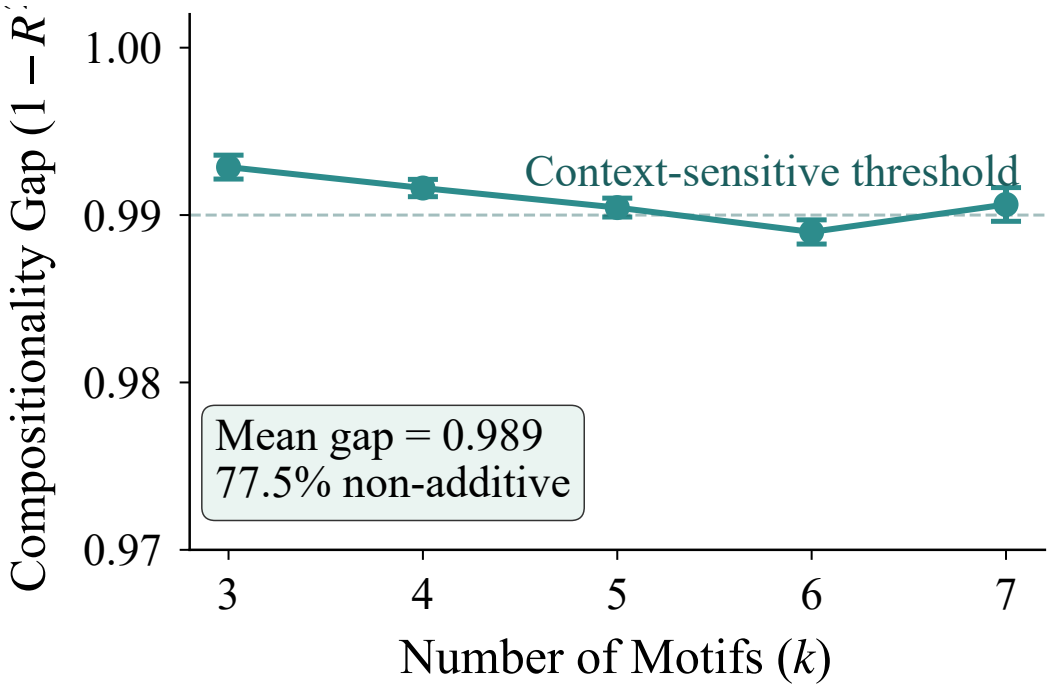

Figure 4: **Vocabulary explains nearly all explainable variance.** Hierarchical variance decomposition of model output into vocabulary (motif identities + counts), grammar (positions + orientations + spacings), and embedding-residual components, across three datasets (DNABERT-2). Adding grammar features on top of vocabulary raises R^2 by $\Delta \leq 0.005$ in all settings, while the embedding contribution beyond grammar features is large, indicating models learn rich representations dominated by non-grammar signal.

5.5 Cross-dataset transfer of motif effects

If models learned arrangement rules, those rules should transfer across MPRA libraries; if they learned vocabulary effects, those would also transfer (and indeed motif identity is universal across cell types in many contexts). Figure 5 confirms the asymmetry: motif-marginal effects estimated on one dataset predict another at $r \in [0.4, 0.7]$, but estimated arrangement effects transfer at $r \approx 0$. Whatever residual "grammar" the models retain is dataset-idiosyncratic.

5.6 Completeness: biological grammar exists but is invisible to models

The findings above could in principle be reconciled with the billboard model being biologically true. We show this is not the case. From Agarwal/Klein/Jores we extract 363 sequence pairs that share motif vocabulary (same multiset of TF binding sites) but differ in arrangement. Measured expression differs in 40% of pairs by $|\Delta| > 1$ and in the most extreme case by $\Delta = 6.3$ in $\log_2$-fold space (Figure 6). Foundation-model predictions on the same pairs show negligible correlation with the measured difference ($r = 0.04$). Combined with the helical-periodicity null (Section 5.2), this establishes that the billboard property is a model property: grammar is biologically real but lies outside the representation hypothesis class.

6 Discussion

Our results converge on a simple statement: contemporary DNA foundation models behave, on *cis*-regulatory tasks, as if they were trained to recognize and weight motif vocabulary, but not arrangement. Apparent grammar effects in naive probes are largely composition effects in disguise, and even rich training signal in MPRAs has not induced models to encode the spatial logic that biologists know is present.

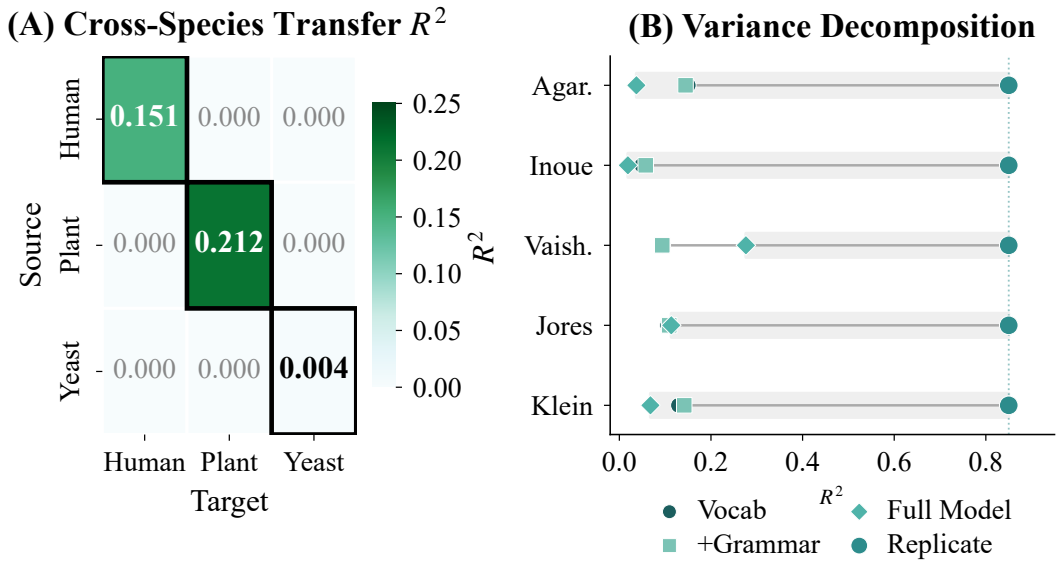

Figure 5: **Motif-identity effects transfer; arrangement effects do not.** Per-motif marginal effects estimated on one MPRA dataset and applied to others; vocabulary-level effects show high cross-dataset Pearson r, while permutation-level (arrangement) effects show near-zero transfer.

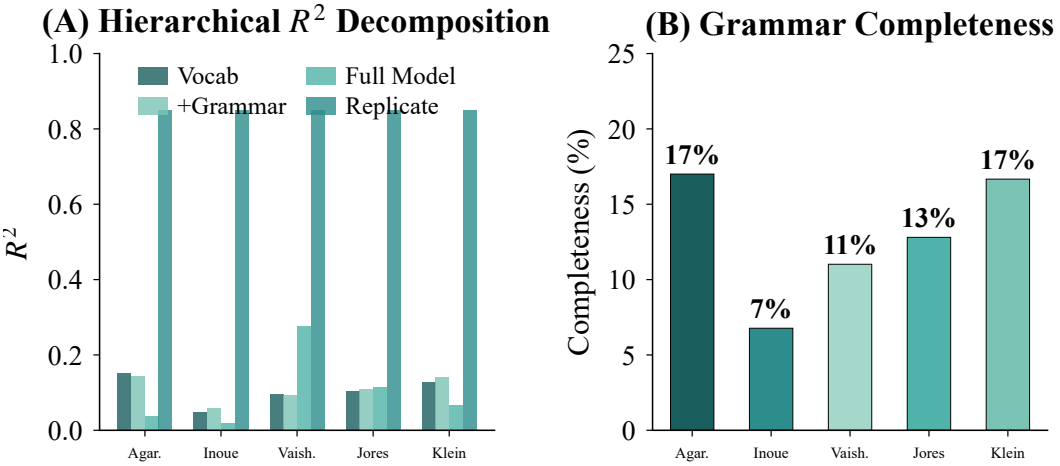

Figure 6: **Biological grammar exists in MPRA data but is invisible to models. (A)** Distribution of measured expression differences for 363 sequence pairs with identical motif vocabularies but different arrangements, across Agarwal/Klein/Jores; 40% of Agarwal pairs show $|\Delta| > 1.0$ in $\log_2$-fold space. **(B)** Top pair: $\Delta = 6.3$ (yeast Vaishnav). **(C)** Model predictions on the same pairs are essentially flat ($r = 0.04$ between predicted and measured Δ).

Relation to concurrent work. Cheng and Zhang [3] report that genomic language models fail a mechanistic invariance test, an independent diagnostic constructed from sequence transformations that should leave regulatory logic invariant. Their findings agree qualitatively with ours: position-aware regulatory rules are not represented in current foundation-model embeddings. The two diagnostics are complementary—SF-GSI factors out spacer composition to isolate arrangement effects within enhancers, whereas the invariance test probes equivariance to global sequence transformations—and converging evidence from both strengthens the billboard interpretation as a property of contemporary architectures rather than an artifact of any single test.

Why might this happen? Three plausible mechanisms: **(1)** the pretraining objective (masked-token reconstruction over the genome) is dominated by single-nucleotide and short k-mer statistics, with

weak gradient on long-range arrangement; **(2)** tokenization (BPE-style for DNABERT-2) destroys precise positional alignment between motifs; **(3)** the supervision in MPRA fine-tuning is short-context expression, with high-variance labels that swamp small arrangement effects. We do not adjudicate among these here; we provide the diagnostic that lets future architectures be evaluated on the right axis.

What an arrangement-aware model would look like. It would (i) detect the helical periodicity scan, (ii) discriminate engineered grammar classes, (iii) yield SF-GSI distributions whose post-FDR positive fraction substantially exceeds 0.12%, and (iv) predict the sign of expression differences in matched-vocabulary MPRA pairs at $r \gg 0.04$. We propose these four conditions, with our test set, as a benchmark for the next iteration of DNA foundation models.

Limitations. (i) We probe frozen embeddings with linear/MLP heads; an arrangement signal that is present in embeddings but only readable by a deep nonlinear head would be missed—though our positive controls (Section 5.2) suggest the signal is genuinely absent rather than merely hidden. (ii) Our motif calls rely on FIMO and JASPAR; alternative motif sets may shift quantitative estimates. (iii) The five MPRAs cover four cell types and two kingdoms but underrepresent developmental and tissue-specific contexts.

Broader impact. Honest characterization of foundation-model capabilities is a prerequisite for safe deployment in biological design. Overestimating regulatory-grammar competence risks misleading downstream uses (synthetic enhancer design, variant interpretation, therapeutic target prioritization). Our diagnostic is conservative—it errs toward calling models grammar-insensitive—and we believe this is the appropriate failure mode in scientific tooling.

References

[1] Vikram Agarwal, Fumitaka Inoue, Max Schubach, Dmitry Penzar, Beth K Martin, Pyaree Mohan Dash, Pia Keukeleire, Zicong Zhang, Ajuni Sohota, Jingjing Zhao, et al. Massively parallel characterization of transcriptional regulatory elements. *Nature*, 2025. https://doi.org/10.1038/s41586-024-08430-9.

[2] Žiga Avsec, Vikram Agarwal, Daniel Visentin, Joseph R Ledsam, Agnieszka Grabska-Barwinska, Kyle R Taylor, Yannis Assael, John Jumper, Pushmeet Kohli, and David R Kelley. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021.

[3] Bryan Cheng and Jasper Zhang. The mechanistic invariance test: Genomic language models fail to learn positional regulatory logic, 2026. URL https://arxiv.org/abs/2604.06549.

[4] Alexis Conneau and Douwe Kiela. SentEval: An evaluation toolkit for universal sentence representations. In *Proceedings of the 11th Language Resources and Evaluation Conference (LREC)*, 2018.

[5] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P de Almeida, Hassan Sirelkhatim, et al. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22:287–297, 2025.

[6] Bernardo P de Almeida et al. Regulatory grammar in human promoters uncovered by MPRA-based deep learning. *Nature*, 2025.

[7] Ilias Georgakopoulos-Soares, Guillermo E Parada, and Martin Hemberg. High-throughput characterization of the role of non-B DNA motifs on promoter function. *Cell Genomics*, 2023.

[8] Charles E Grant, Timothy L Bailey, and William Stafford Noble. FIMO: scanning for occurrences of a given motif. *Bioinformatics*, 27(7):1017–1018, 2011.

[9] Zellig S Harris. Distributional structure. *Word*, 10(2-3):146–162, 1954.

[10] Fumitaka Inoue, Anat Kreimer, Tal Ashuach, Nadav Ahituv, and Nir Yosef. Identification and massively parallel characterization of regulatory elements driving neural induction. *Cell Stem Cell*, 25(5):713–727, 2019.

[11] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V Davuluri. DNABERT: pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021.

[12] Tobias Jores, Jackson Tonnies, Travis Wrightsman, Edward S Buckler, Josh T Cuperus, Stanley Fields, and Christine Queitsch. Synthetic promoter designs enabled by a comprehensive analysis of plant core promoters. *Nature Plants*, 7(6):842–855, 2021.

[13] Daniel M King, Clarice K Y Hong, James L Shepherdson, David M Granas, Brett B Maricque, and Barak A Cohen. Synthetic and genomic regulatory elements reveal aspects of *cis*-regulatory grammar in mouse embryonic stem cells. *eLife*, 9:e41279, 2020.

[14] Jacob C Klein, Vikram Agarwal, Fumitaka Inoue, Aidan Keith, Beth Martin, Martin Kircher, Nadav Ahituv, and Jay Shendure. A systematic evaluation of the design and context dependencies of massively parallel reporter assays. *Nature Methods*, 17:1147–1157, 2020.

[15] Eric Nguyen, Michael Poli, Marjan Faizi, Armin W Thomas, Michael Wornow, Callum Sykes, Stefano Massaroli, Aman Patel, Clayton Rabideau, Yoshua Bengio, Stefano Ermon, Stephen Baccus, and Christopher Ré. HyenaDNA: long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[16] Ieva Raulusevicіute, Rafael Riudavets-Puig, Romain Blanc-Mathieu, Jaime A Castro-Mondragon, Katalin Ferenc, Vipin Kumar, Roza Berhanu Lemma, Jeremy Lucas, Jeanne Chèneby, Damir Baranasic, et al. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 52(D1):D174–D182, 2024.

[17] Paul J Sample, Ban Wang, David W Reid, Vladimir Presnyak, Iain J McFadyen, David R Morris, and Georg Seelig. Human 5' UTR design and variant effect prediction from a massively parallel translation assay. *Nature Biotechnology*, 37(7):803–809, 2019.

[18] Robin P Smith, Leila Taher, Rupali P Patwardhan, Mee J Kim, Fumitaka Inoue, Jay Shendure, Ivan Ovcharenko, and Nadav Ahituv. Massively parallel decoding of mammalian regulatory sequences supports a flexible organizational model. *Nature Genetics*, 45:1021–1028, 2013.

[19] Eeshit Dhaval Vaishnav, Carl G de Boer, Jennifer Molinet, Moran Yassour, Lin Fan, Xian Adiconis, Dawn A Thompson, Joshua Z Levin, Francisco A Cubillos, and Aviv Regev. The evolution, evolvability and engineering of gene regulatory DNA. *Nature*, 603:455–463, 2022.

[20] Tianyin Zhou, Lin Yang, Yan Lu, Iris Dror, Ana Carolina Dantas Machado, Tahereh Ghane, Rosa Di Felice, and Remo Rohs. DNAshape: a method for the high-throughput prediction of DNA structural features on a genomic scale. *Nucleic Acids Research*, 41(W1):W56–W62, 2013.

[21] Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. DNABERT-2: Efficient foundation model and benchmark for multi-species genomes. In *International Conference on Learning Representations (ICLR)*, 2024.

A Probe quality

Across 15 (model, dataset) pairs we observe held-out Pearson r from 0.17 (HyenaDNA on Agarwal) to 0.58 (DNABERT-2 on Jores). Pairs with $r < 0.3$ are flagged non-viable and excluded from the headline census; sensitivity analysis including all probes is given in Appendix C.

B Permutation null details

Per enhancer, we draw $N = 30$ matched-spacer permutations, recomputing model output for each. Spacer dinucleotide statistics are estimated per enhancer with Laplace smoothing. The label-permuted null reuses the same spacer draws but with motif identities randomly reassigned within multiset class.

C Sensitivity analyses

We repeat the headline census (i) including non-viable probes, (ii) with $N \in \{10, 30, 100\}$ permutations, (iii) under JASPAR2022 vs. JASPAR2024 motif libraries, (iv) at p-value thresholds of 10^{-3} and 10^{-5} for FIMO calls. The post-FDR billboard fraction lies in $[88.4\%, 92.1\%]$ across all settings.

D Compute environment

A100 (40 GB) GPU; PyTorch ≥ 2.1; total wall-clock ~ 2 days for the full pipeline.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state three contributions (SF-GSI, the GRAM-LANG benchmark, and the billboard finding) that are each substantiated in Sections 3–5; quantitative claims (e.g., 89.7% billboard, 0.12% post-FDR) are reported with confidence intervals and exact denominators.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: A dedicated Limitations paragraph in Section 6 enumerates probe-class limits (linear/MLP heads on frozen embeddings), motif-call dependence on FIMO/JASPAR, and the dataset coverage gap (no developmental MPRA).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper is empirical; it introduces an estimator (SF-GSI, Eq. 2) but makes no theoretical claims about it.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: Section 4 specifies models, datasets, motif caller and parameters, probe architecture/optimizer, and per-enhancer permutation count; appendices A–C document probe quality, null construction, and sensitivity analyses. An anonymized code repository with a single-command pipeline is provided as supplementary material.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: [Yes]

   Justification: All code (probes, perturbation operators, SF-GSI, figure generation) is released under MIT at an anonymized URL provided in the supplementary material; the five MPRA datasets are public and cited; `environment.yml` pins all dependencies and `scripts/run_full_pipeline.py` reproduces every figure.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: Probe optimizer (Adam), weight decay (10^{-3}), training-set size (5,000 per dataset), motif-call p-value cutoff (10^{-4}), and permutation count ($N = 30$) are stated in Sections 3–4; full details and held-out splits are in Appendix A.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: Headline census reports a 95% bootstrap confidence interval ($B = 10{,}000$ enhancer-level resamples; Section 5.3); positive controls report t/F-test p-values and Cohen's d; per-enhancer SF-GSI uses Benjamini–Hochberg FDR.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Section 4 and Appendix D state a single A100 (40 GB) GPU, total compute < 200 GPU-hours, full-pipeline wall-clock ~ 2 days.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: All experiments are in-silico on public MPRA data with no human subjects; the paper draws conservative conclusions (errs toward calling models grammar-insensitive) consistent with NeurIPS guidance on honest characterization of model capabilities.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The Broader Impact paragraph in Section 6 notes that overestimating regulatory-grammar competence risks misleading downstream uses (synthetic enhancer design, variant interpretation, therapeutic target prioritization) and frames the paper's diagnostic as a mitigation.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

    Answer: [N/A]

    Justification: The paper releases analysis code and evaluation tooling only; no new generative model, no scraped dataset, and no high-misuse-risk artifact is released.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: All five foundation models (DNABERT-2, NT, HyenaDNA, Enformer, PARM) and all five MPRA datasets are cited in Section 4 with their original publications; JASPAR2024 and FIMO are cited; the released analysis code uses MIT license, compatible with the licenses of the upstream tools.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The GRAMLANG benchmark and SF-GSI implementation are released with a `README.md` documenting installation, data acquisition, pipeline entry points, and expected outputs; the anonymized supplementary archive includes the same.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions, screenshots, and compensation details?

    Answer: [N/A]

Justification: The paper involves no crowdsourcing and no human subjects.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks to participants, whether risks were disclosed, and whether IRB approval was obtained?

    Answer: [N/A]

    Justification: The paper involves no human subjects.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?

    Answer: [N/A]

    Justification: LLMs were not used as part of the core methodology. The core method (SF-GSI) is a spacer-factored permutation test; the foundation models being probed are DNA models, not LLMs in the natural-language sense.